AYMAN MIDAN

Engineering Student — Machine Learning & Artificial Intelligence

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PROFILE

Engineering student in Machine Learning & AI, ranked 1st/39 in most recent semester. I design and deploy deep learning, generative AI, and computer vision solutions for real-world problems. Rigorous and comfortable with both implementation and analysis, I am looking to contribute to ambitious, high-impact technical projects.

EDUCATION

Grenoble INP – ENSIMAG — 3rd Year, Information Systems Engineering, ML & AI Track — 2026 – present

INPT Rabat — Engineering Cycle, Computer Science — 2024 – 2026

CPGE MP — Lycée d'Excellence Moulay Youssef, Rabat — 2022 – 2024

PROFESSIONAL EXPERIENCE

Research Internship — Robust Aerial Object Detection for Drones and eVTOLs — Jul. – Aug. 2026
Sejong University, Seoul, South Korea — Remote

- Obstacle detection under degraded visibility conditions using **YOLOv8** and deep learning.

Internship — Intelligent Password Dictionary Generation via GANs — Jun. – Jul. 2026
Scientiae Rabat — Hybrid Internship

- Developed a password generator using **SeqGAN** with an **LSTM** generator trained via *policy gradient*.
- Evaluated generated distributions, analyzing diversity and coverage of produced dictionaries.

Researcher — Fairness-Aware Federated Unlearning for IoT Networks — Mar. – Jun. 2026
INPT Rabat — Collaboration with PhD Students

- Studied selective unlearning within a federated learning framework under fairness constraints across IoT nodes.
- Implemented on the ML-EdgeIIoT dataset with Non-IID FedAvg and Knowledge Distillation.

PROJECTS

Adversarial ML for Intrusion Detection (IDS) — Python, PyTorch, Scikit-Learn

- Physically-constrained attacks (CPGD, PSO) on an MLP-based IDS — **+90% black-box evasion rate**.
- Adversarial training reducing white-box evasion rate to **2.4%**.

Ransomware Detection via Generative AI — Python, GANs, VAEs, CTGAN

- Data augmentation via **GANs** and **VAEs** to improve detection of minority classes.
- **94% accuracy** on 100,000 CTGAN-generated synthetic samples.

Local RAG System for PDF Documents — LangChain, ChromaDB, FastAPI, Streamlit, Ollama

- Full pipeline: ingestion → embeddings → vector search → context-aware generation.
- Conversational memory and source attribution via an interactive web interface (Streamlit).

TECHNICAL SKILLS

Machine Learning / Deep Learning: Scikit-learn, PyTorch, TensorFlow, MLP, GANs, VAEs

LLM / Generative AI: LLMs, Prompt Engineering, LangChain, RAG, Embeddings, Fine-tuning

Computer Vision: OpenCV, YOLOv8

Data Engineering: Pandas, NumPy, Preprocessing, Feature Engineering, Data Augmentation

Backend / Deployment: Git, FastAPI, Streamlit, Docker, REST API

Databases: SQL, MongoDB, ChromaDB

Languages: Python, Java, SQL, Bash

CERTIFICATIONS

Kaggle — Machine Learning | Kaggle — Computer Vision | Oracle — Generative AI (in progress)

LANGUAGES

Tamazight (native) | Arabic (professional) | French (professional) | English (professional) | German (basic) | Spanish (basic)